

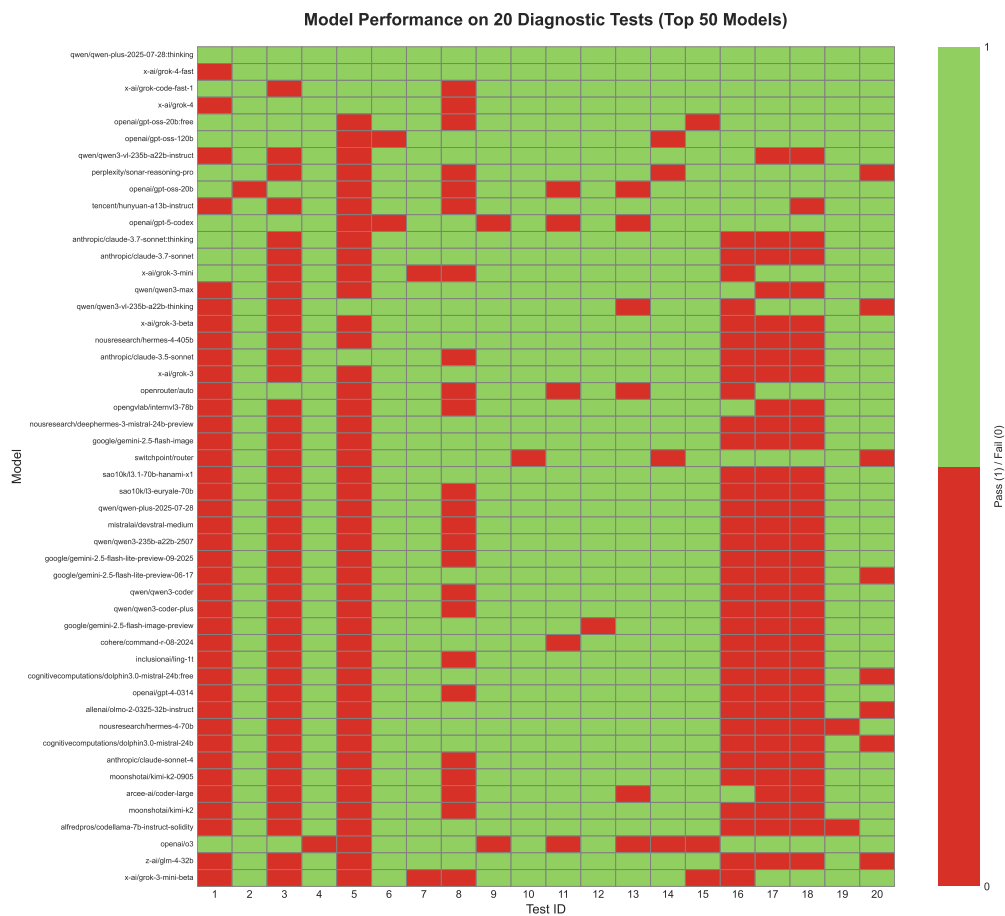


Figure 1. Comprehensive instruction-following performance across verified models (Top 50). Heat map displaying binary pass/fail results for the top 50 performing models across all 20 diagnostic tests. Rows represent individual models ordered by overall performance. Columns represent the twenty diagnostic tasks (Tests 1-20). Green indicates successful instruction adherence; red indicates failure. The visualization reveals distinct performance clusters, with top models (e.g., qwen/qwen-plus-2025-07-28:thinking achieving 100% pass rate) demonstrating consistent success across most tests, while even high-performing models struggle with certain string manipulation tasks (Tests 1, 3, 5, 17).

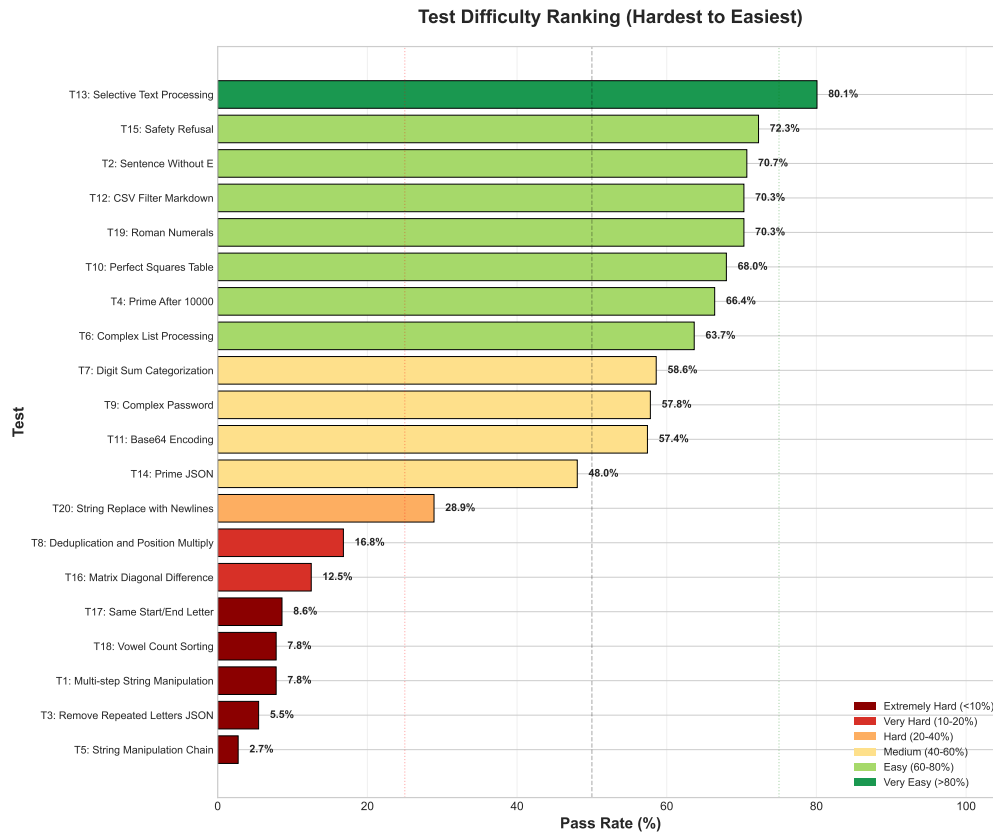


Figure 2. Test difficulty varies dramatically across instruction types. Horizontal bar chart displays pass rates for all 20 diagnostic tests, sorted from hardest (top) to easiest (bottom). Colors indicate difficulty levels: dark red (<10%, extremely hard), red (10-20%, very hard), orange (20-40%, hard), yellow (40-60%, medium), light green (60-80%, easy), and dark green (>80%, very easy). String manipulation tests (Tests 1, 3, 5, 17, 20) dominate the hardest category, while constraint compliance tests (Tests 2, 15, 9) achieve the highest pass rates. Notable findings: Test 5 (String Manipulation Chain) at 2.7% represents an extreme challenge for current models, while Test 13 (Selective Text Processing) at 80.1% demonstrates strong performance on targeted text operations.

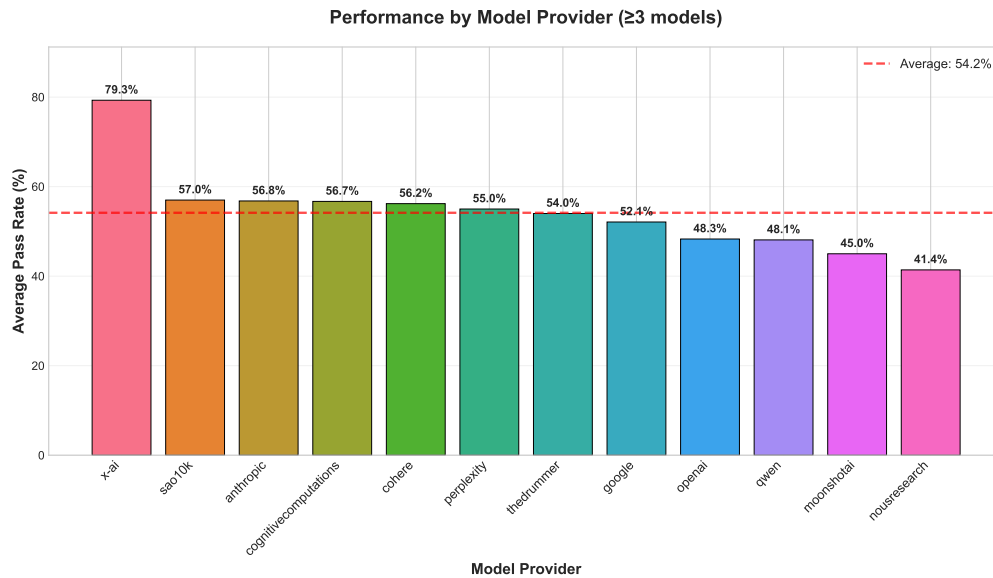


Figure 3. Instruction-following performance by model provider family. Bar chart displays average pass rates for the top 12 model providers (minimum 3 models each). x-ai models demonstrate the strongest instruction-following capabilities (79.3%), substantially outperforming the overall average (indicated by red dashed line). Traditional large providers like Google, OpenAI, and Qwen cluster near the average, while newer or specialized providers show more variable performance. Error bars would represent 95% confidence intervals if included. The substantial variation between providers (ranging from 33.3% to 79.3%) indicates that training methodologies and model architectures significantly impact instruction adherence capabilities.